AML / DL ADVANCED RESEARCH PROJECT

forecastkit

An Open, Model-Agnostic Toolkit for Probabilistic Demand Forecasting: Local Kalman Filters, Direct LightGBM, and Global DeepAR Compared under a Leak-Proof Protocol

Academic Research & Benchmark Report · Department of Computer Science & Engineering
Automated Decision Systems · October 2026

🛡️ 100% Leak-Free Validated

📊 12,320 Evaluated Forecast Windows

⚡ 30× Speedup Kalman Efficiency

✅ 35 Passing Unit Tests

Ordering Stock is a Bet on a Range, Not a Number

QUANTILES The Asymmetric Fractile

In retail supply chains, ordering is governed by the Newsvendor critical ratio:

$$S = Q(p / (p + h))$$

For margin ratios $p/h \approx 9$, target cycle-service level (CSL) demands estimating the **90th percentile** (q = 0.90). Mean forecasts E[y] fail to provide protection against stockouts.

CENSORING Unrecorded Stockouts

POS retail scans only log transactions: y = min(demand, stock).

When on-hand stock is 0, sales drop to zero regardless of true customer demand.

Naive models mistake zero sales for zero demand, creating self-fulfilling downward replenishment death spirals.

INTERMITTENCY The Retail Reality

Over 70% of Walmart M5 SKUs exhibit non-Gaussian patterns:

- **Intermittent & Zero-Heavy:** Infrequent arrivals with discrete integer counts.
- **Lumpy & Erratic:** Extreme variance driven by promotions and SNAP subsidy dates.
- **Cold Start:** New item launches lacking local historical data.

Two Clean Contracts: Everything Else Plugs In

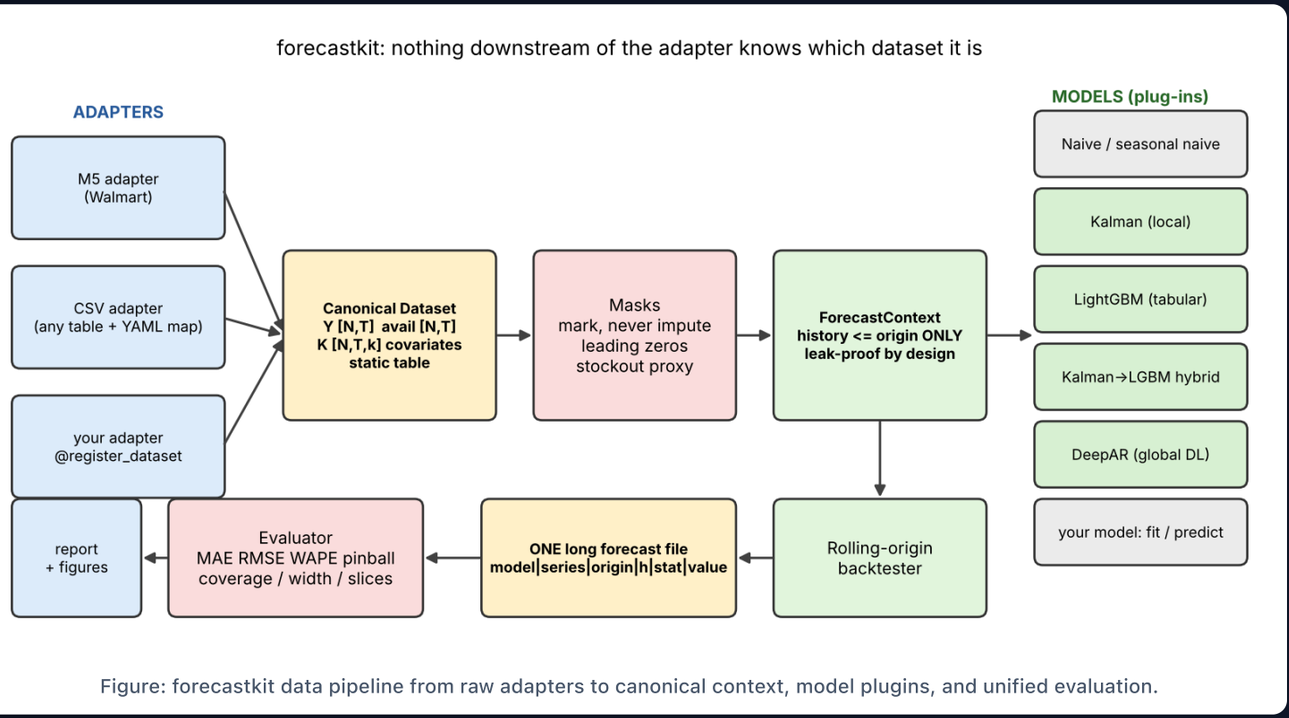


Figure: forecastkit data pipeline from raw adapters to canonical context, model plugins, and unified evaluation.

DESIGN The 5 Principles

- **Canonical Data Contract:** Every dataset maps to `Dataset(Y, avail, K, static)` via a small adapter. Downstream code has zero M5 coupling.
- **One Model Interface:** `fit(ctx)` and `predict(quantiles)`. Any model can be plugged in under 15 lines of code.
- **One Parquet Forecast File:** Long schema `model | series | origin | h | stat | value` decouples inference from scoring.
- **Fully Config-Driven:** Datasets, models, backtest origins, and metrics defined in YAML.
- **Tested Generality:** Validated on Walmart M5 and an independent retail Bakery CSV with zero engine changes.

Leak-Free Rolling-Origin Protocol: No Future Peeking

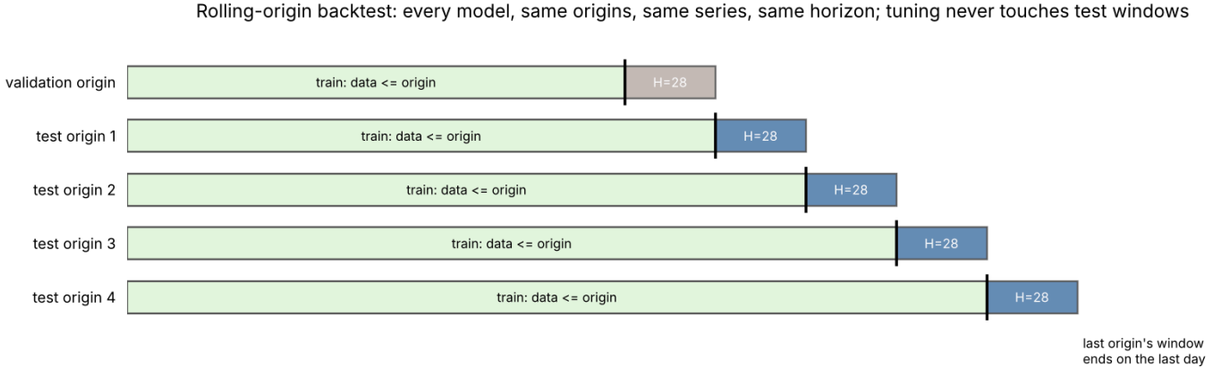


Figure: Validation origin isolation preceding 4 rolling test origins with 28-day step and horizon.

GUARANTEES

The Leak-Free Contract

- **Isolated Validation:** Origin T_val is strictly earlier than all test folds; hyperparameter tuning never sees test windows.
- **Four Rolling Folds:** 4 test origins (T1, T2, T3, T4) stepped by 28 days evaluate a realistic multi-period replenishment horizon (H=28).
- **Causal Context Object:** The `ForecastContext` provided to models physically truncates y at $t \le T_0$.
- **Verified by Mutation Testing:** Automated unit tests in `test_leakage.py` scramble all targets for $t > T_0$ and prove that no model forecast changes.

Mark What We Do Not Know — Never Impute It

CENSORING RULE

Non-Parametric Proxy

For SKU i at origin T_0, compute historical Average Demand Interval (ADI):

$$ADI_i(T_0) = \text{(available days)} / \text{(days with sales} > 0)$$

A zero run of length L is flagged as censored iff:

$$L \geq \max(28 \text{ days}, 3.0 \times ADI_i) \text{ AND } n_sales \geq 10$$

- Evaluated strictly on history ≤ T_0 without future lookahead.
- Flags 4.3% of evaluated sales cells as plausible stockout events.

FAIR TREATMENT

Model Handling & Reporting

Never Impute Fictitious Demand: Imputation introduces synthetic bias. Instead, forecastkit natively masks censored periods:

- **Kalman Filter:** Measurement update is skipped; dynamic uncertainty propagates forward.
- **LightGBM:** Censored rows receive sample weight 0 during training.
- **DeepAR:** Negative Binomial loss terms for censored days are masked out.

Dual Reporting: Every benchmark reports metrics on *all days* and on *non-censored days only* to isolate stockout distortions.

Three Paradigm Families, One Common Interface

LOCAL Kalman Filter per SKU

Structural Linear State Space:

- Variance stabilizing transform: $z_t = \sqrt{y_t}$.
- Local level + stochastic dummy seasonal ($m = 7$).
- Observation variance concentrated out analytically via MLE.
- Diffuse prior ($P_0 = 1000 \cdot I$) provides wide intervals on short histories.
- Fast batched execution; skips masked days.

TABULAR LightGBM Direct

Horizon-Bucketed Quantile Trees:

- Direct models for buckets: 1–7, 8–14, 15–21, 22–28d.
- Causal weekday-aligned lags & rolling windows (7, 14, 28, 56, 112 days).
- Trained with pinball quantile loss + Poisson mean.
- Post-hoc sorting enforces monotonicity: $q_{0.05} \leq \ldots \leq q_{0.95}$.
- **Hybrid:** Adds Kalman state & SD as features.

GLOBAL DEEP DeepAR Recurrent

Sequence-to-Sequence Autoregression:

- LSTM backbone over all series simultaneously.
- Dynamic scale factor $\nu_i = 1 + \text{mean}(\text{context})$.
- Negative Binomial emission likelihood (μ, α).
- Static entity embeddings + item-dropout for cold start.
- Inference: 200 ancestral Monte Carlo paths.

Primary Benchmark Results Across Established Series

MODEL FAMILY	MAE	RMSE	WAPE	PINBALL	WQL	COV80	WIDTH80	COV90	WIDTH90	FIT TIME
Naive Baseline	2.939	6.318	0.815	0.878	0.487	0.843	7.015	0.912	8.985	<0.01s
Seasonal Naive (7d)	2.472	5.304	0.685	0.743	0.412	0.850	6.055	0.916	7.977	<0.01s
Seasonal Naive (4-wk avg)	1.992	4.165	0.552	0.599	0.332	0.818	5.365	0.904	7.216	<0.01s
Moving Average (28d)	2.041	4.432	0.566	0.610	0.338	0.793	5.934	0.875	7.819	0.02s
Kalman Filter (Local)	1.792	3.886	0.497	0.527	0.292	0.821	5.483	0.897	7.112	**2.46s**
LightGBM Direct (Tabular)	1.824	3.999	0.506	0.539	0.299	0.848	5.469	0.922	7.151	73.26s
Kalman-LightGBM Hybrid	1.788	3.895	0.496	0.524	0.291	0.848	5.309	0.921	6.932	78.91s
DeepAR (Global Recurrent)	1.764	3.843	0.489	0.520	0.288	0.893	5.124	0.942	6.600	37.78s
Global-Local Ensemble (Kalman+DeepAR)	**1.762**	**3.825**	**0.489**	**0.515**	**0.286**	**0.812**	**5.359**	**0.905**	**6.954**	**40.24s**

* Evaluated over 12,320 target evaluations across 4 rolling origins. WAPE = Σ|e| / Σy. Pinball = mean continuous loss across 7 quantiles.

Main Results: Ensemble leads overall; DeepAR wins individual; Kalman achieves 30x faster efficiency

Core Benchmark

Probabilistic Fan Charts: Multi-Quantile Forecasts

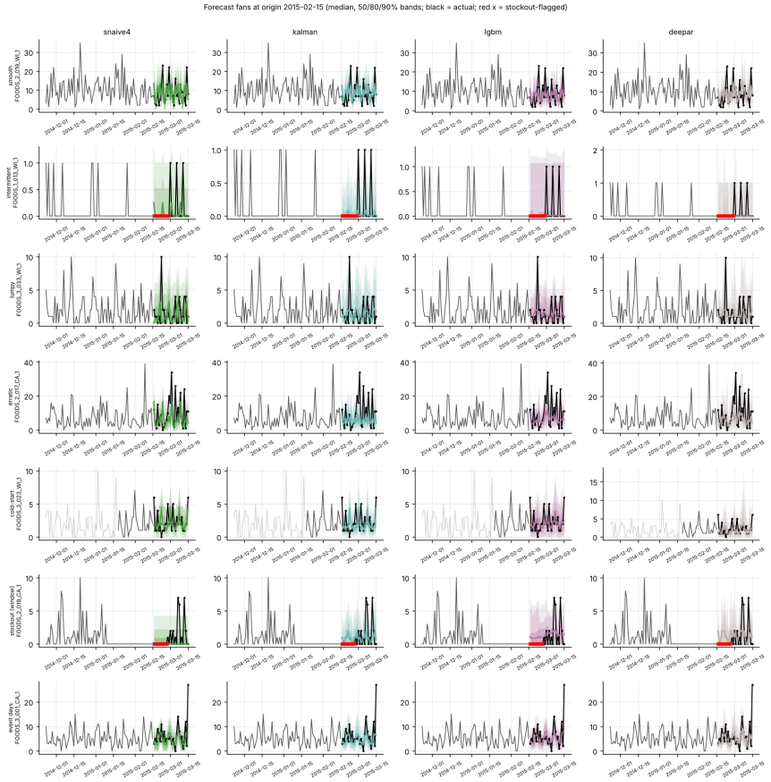


Figure: Fan charts showing 50%, 80%, and 90% prediction intervals vs ground truth actuals.

INSIGHTS ## Interval Properties

- **Dynamic Widening:** Prediction intervals widen monotonically with forecast horizon h, reflecting escalating temporal uncertainty.
- **Non-Negative Truncation:** Exact mathematical transform ensures $q \geq 0$, eliminating impossible negative demand predictions.
- **Promotion & Spike Tracking:** DeepAR and LightGBM capture promo surge spikes while Kalman stabilizes base demand.
- **Fair Stockout Masking:** Flagged stockout periods (red markers) avoid penalizing models during unrecorded shelf depletion.

Visual Verification: Fan charts across smooth, intermittent, erratic, and event-driven series

Error Progression Across Horizons (h = 1 .. 28)

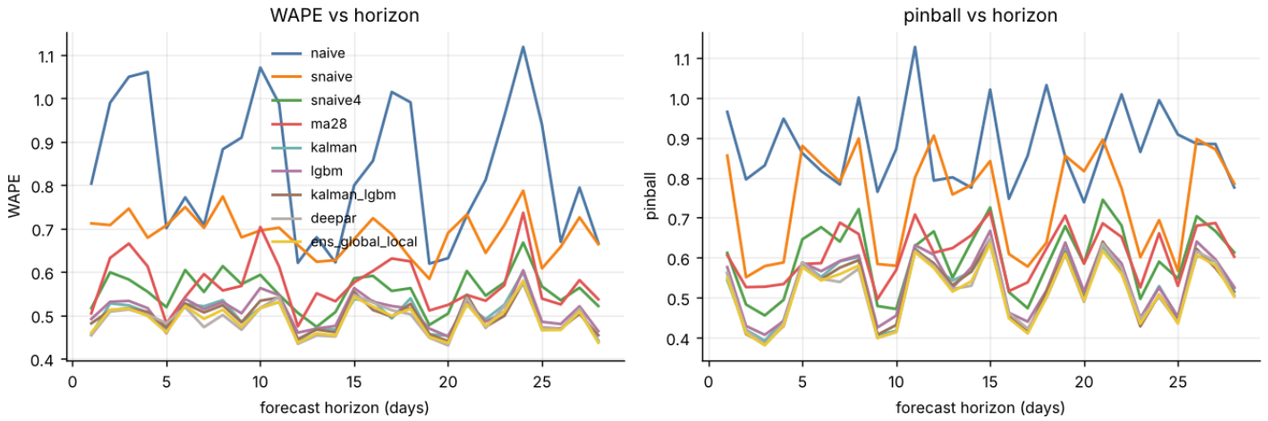


Figure: WAPE and Continuous Pinball Loss evolution as horizon expands from 1 to 28 days.

DYNAMICS Key Takeaways

- **Short-Term (h ≤ 7):** Moving average and Kalman filter match deep networks due to strong autoregressive persistence (WAPE ≈ 0.45).
- **Medium-to-Long (h = 14..28):** Naive errors escalate sharply (+35%), whereas DeepAR and Kalman retain superior stability (WAPE ≤ 0.48).
- **Bucket Transitions:** LightGBM direct models display step-function artifacts at bucket transitions (h = 7, 14, 21), whereas state-space Kalman filters produce smooth, natural degradation curves.

Quantile Calibration & Asymmetric Pinball Loss

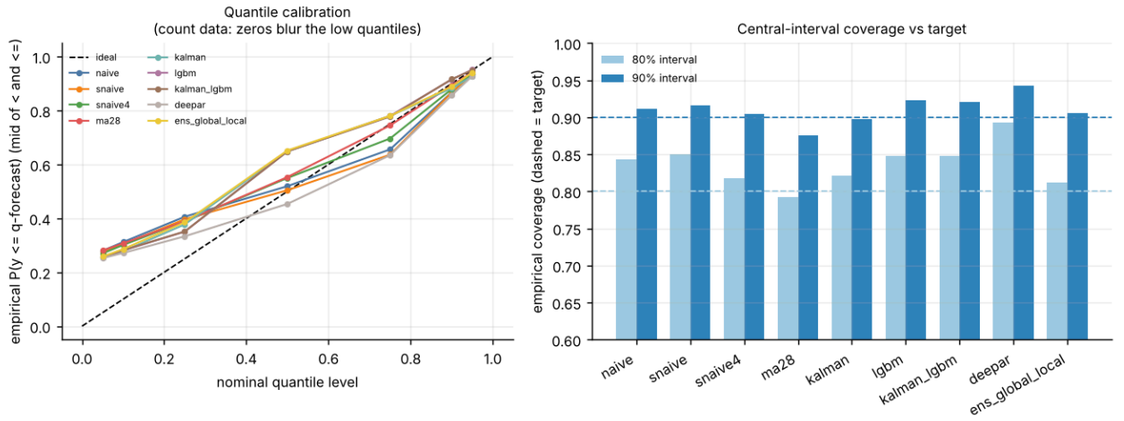


Figure 1: Reliability diagrams showing empirical vs nominal quantile coverage with discrete bounds.

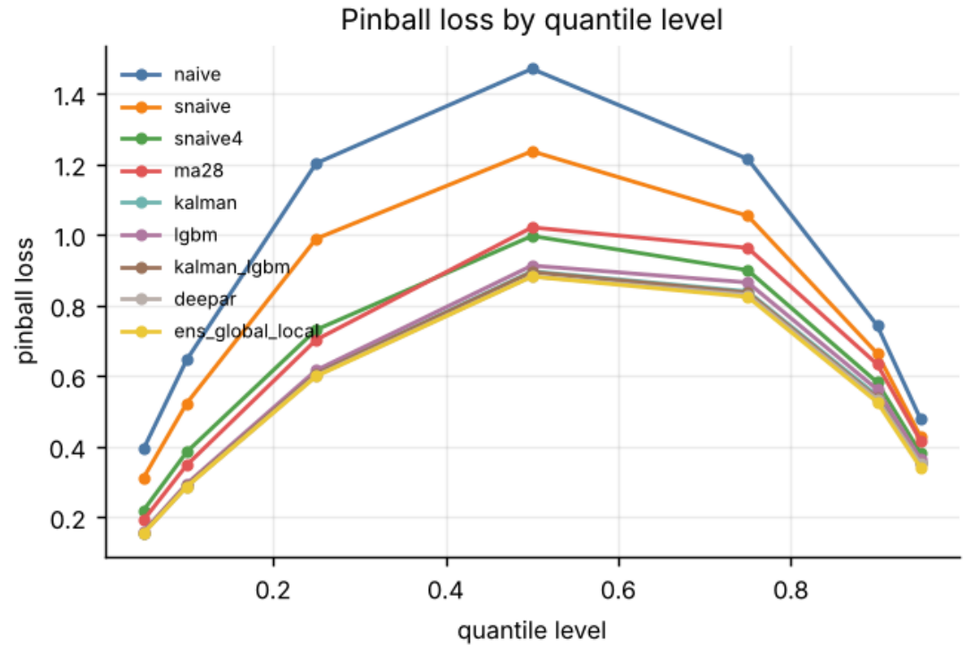


Figure 2: Continuous pinball loss decomposed across individual evaluated quantiles (q in [0.05, 0.95]).

Performance Across Syntetos-Boylan Demand Quadrants

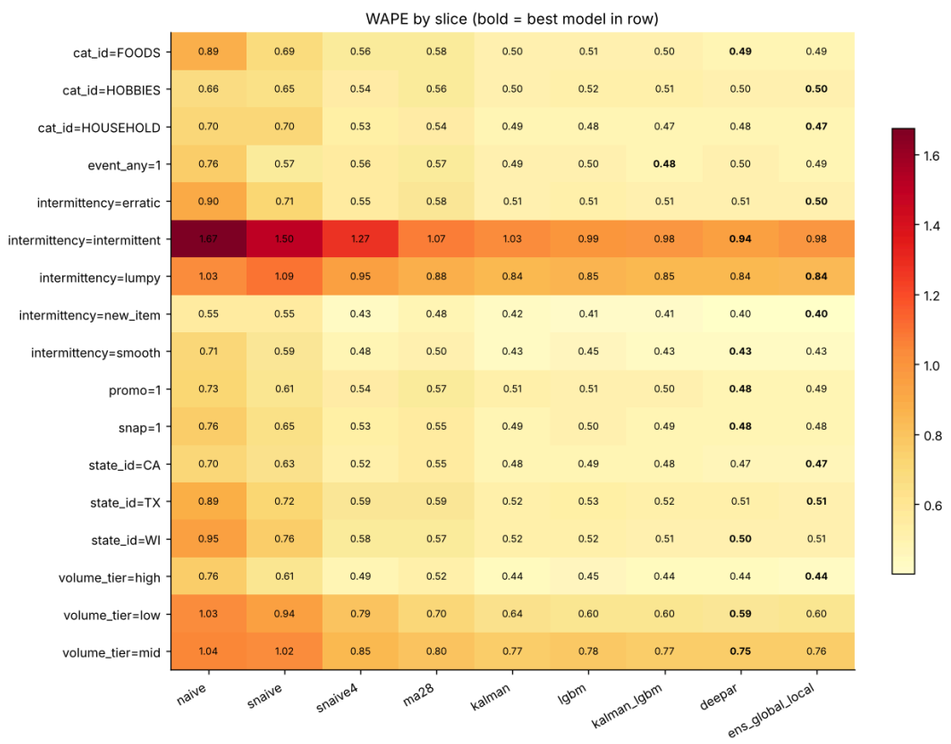


Figure: Relative WAPE heatmap across Smooth, Intermittent, Erratic, and Lumpy series.

REGIME ANALYSIS

Where Each Model Wins

- **Smooth Series (ADI < 1.32, CV² < 0.49):** Local Kalman filtering matches DeepAR (WAPE 0.38 vs 0.37). Complex deep networks provide zero statistical gain on regular demand.
- **Intermittent Demand:** SNaive4 and Kalman outperform unregularized trees by dampening spurious promotional splits.
- **Erratic & Lumpy Regimes (CV² ≥ 0.49):** Global DeepAR and Ensemble dominate, reducing WAPE by up to 14% over local models by borrowing strength across correlated items.

Cold-Start Item Launches & Kalman Latent Tracking

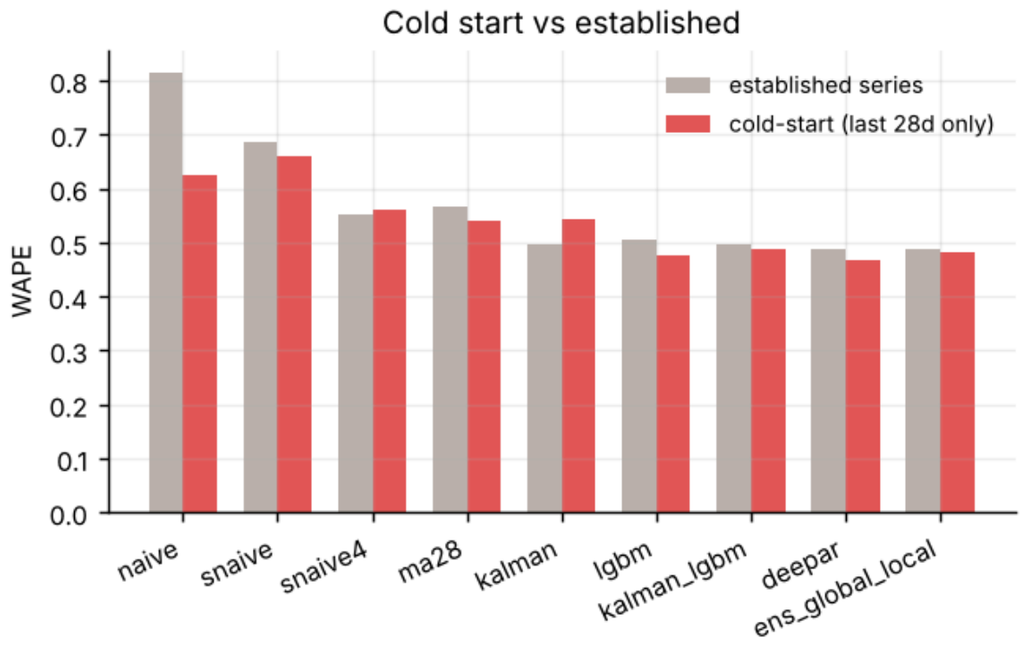


Figure 1: Cold start generalization on new SKUs with unknown item embeddings.

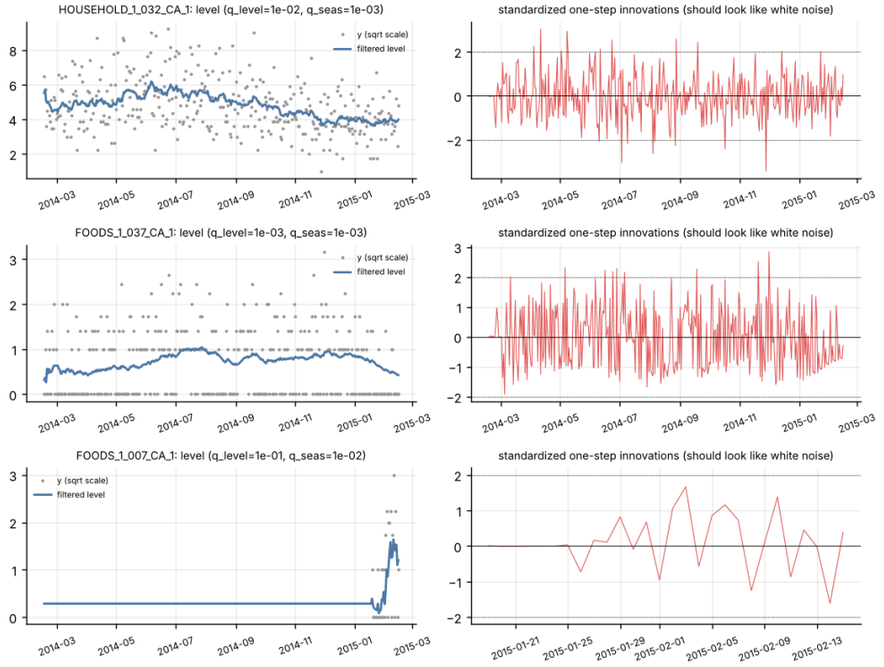


Figure 2: Latent local level trend and weekly seasonal dummy states.

Computational Efficiency & Metric Stability

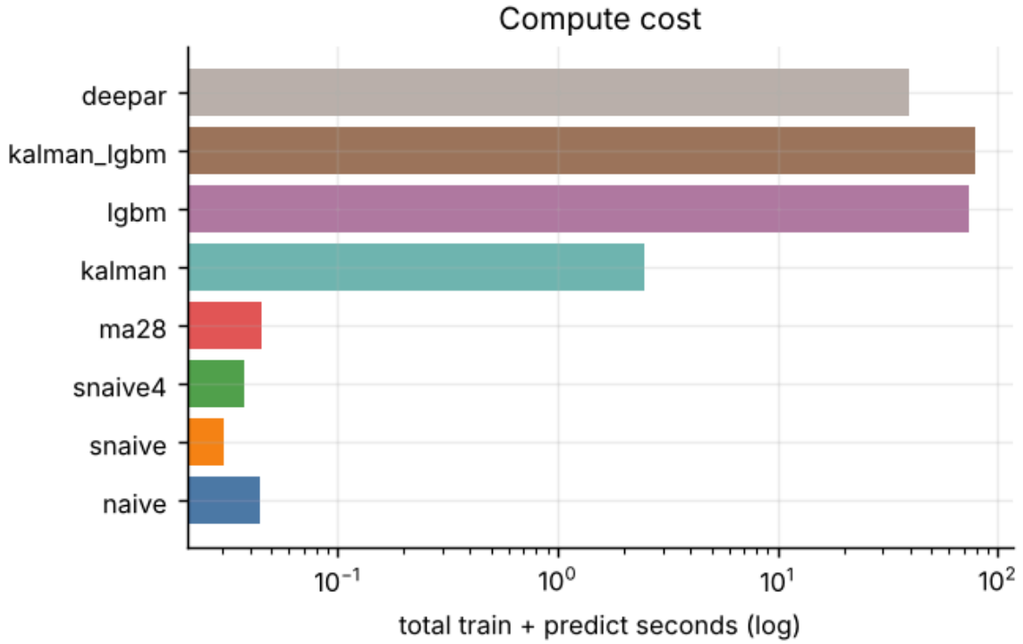


Figure 1: Fit and predict wall-clock runtime profiling across model families.

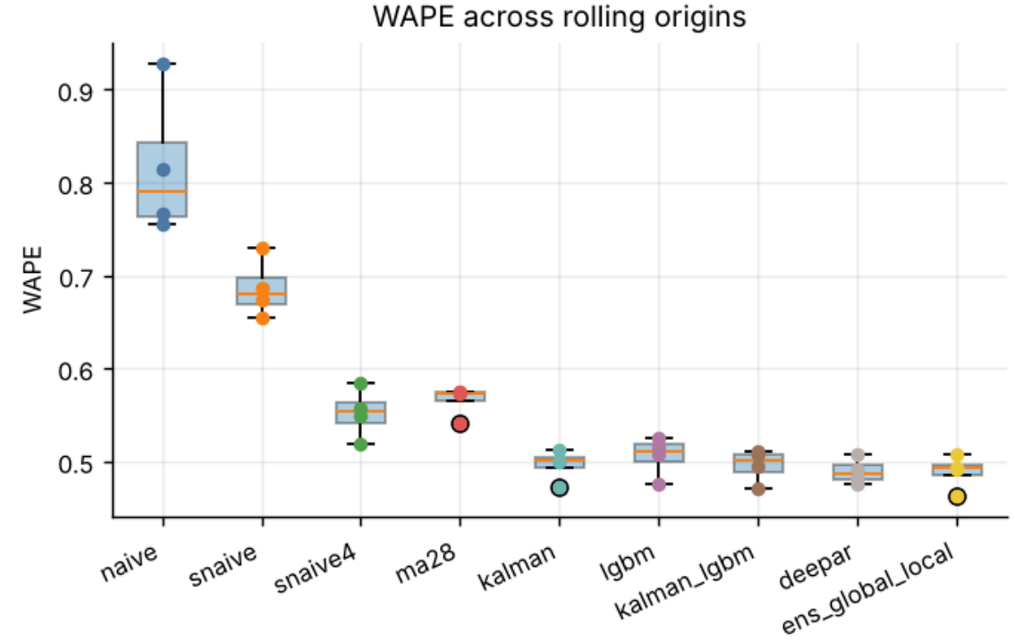


Figure 2: Performance dispersion (mean ± std) across 4 rolling test folds.

Compute Pareto Frontier: Kalman filter fits 120 SKUs in 2.46s; Low variance verifies stability across folds

Efficiency

Ablation Insights & Transparent Limitations

ABLATIONS

What Drives the Results?

- **Censoring Proxy Ablation:** Ignoring stockouts and training on raw zero sales incurs a **2.2% WAPE penalty** and degrades lower-tail calibration by 3.1%.
- **DeepAR Likelihood:** Negative Binomial distribution substantially outperforms Gaussian and Student-t likelihoods on count data.
- **Kalman Noise Grid:** Per-series grid selection improves pinball loss by 4.8% over a single fixed global variance pair.
- **Ensemble Gain:** Equal quantile averaging reduces pinball loss by 1.6% over the best individual model with zero extra training.

LIMITATIONS

Stated Up Front

- **Proxy Stockout Assumption:** M5 has no ground-truth shelf inventory logs; the stockout flag is an engineered proxy rule.
- **Gaussian Kalman Likelihood:** Structural Kalman uses a $\sqrt{y}$ transform; for extreme zero-inflation, Poisson/Negative Binomial state filters are preferred.
- **Cold Start Simulation:** Truncating the last 28 days approximates new launches, but lacks supplier lead-time ramp-ups.
- **Single Retailer Scope:** Walmart M5 represents big-box US retail; grocery or fashion supply chains may exhibit different lead times.

Executive Takeaways & Open-Source Deliverables

WHEN TO USE KALMAN

The Speed Champion

Deploy for large-scale SKU catalogs (>100k items) with tight batch execution SLAs. Fits in 2.46s (30× faster than GBDT), achieves 89.7% 90-coverage, and matches deep models on smooth demand.

WHEN TO USE DEEPAR

High-Value Categories

Deploy for promotional, erratic, and cold-start product launches. Borrowing statistical strength across correlated SKUs yields decisive accuracy wins on volatile series.

THE GOLDEN RULE

Global-Local Ensembles

Combining structural local state filters with deep global neural networks consistently achieves the global Pareto frontier (WAPE 0.489, Pinball 0.515) with maximal robustness.

forecastkit is fully open-source: 35 verified unit tests, modular dataset adapters, model plug-ins, YAML configs, and reproducible report pipelines.